



Course Name

GENERAL PRACTICAL PHYSICS II

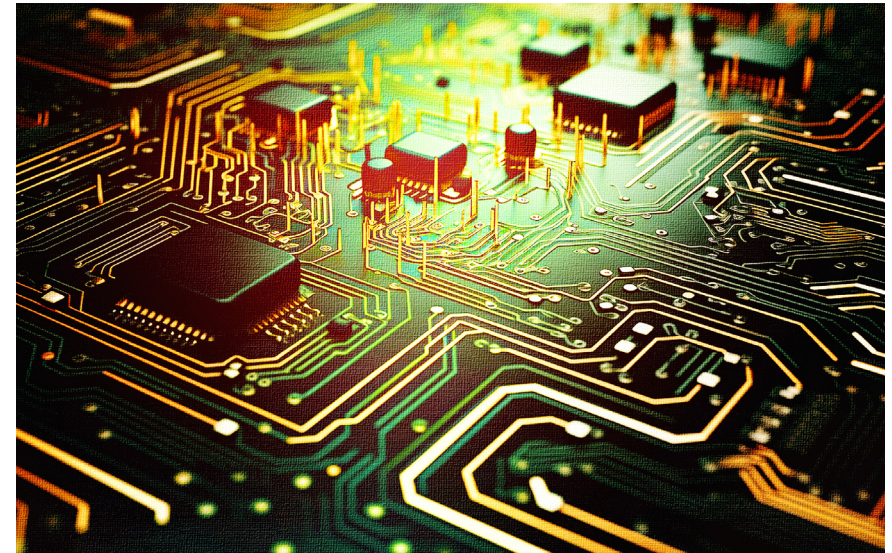
Topic: Experiment to Verify Nonlinear Relationship
Between Current and Voltage in a Non-Ohmic
Resistor

WEEK 3

IN SUMMARY

This week, I learned about non-ohmic resistors, which are resistors that do not obey Ohm's law. This means that their resistance changes as voltage and current vary. Some common examples of non-ohmic resistors include light bulbs, diodes, and transistors. Non-ohmic resistors can exhibit a variety of interesting characteristics, such as non-constant resistance, non-reversible behaviour, and sensitivity to temperature and light. Non-ohmic behaviour is commonly observed in semiconductors and is utilized in devices like thermistors and LDRs.

The key distinction between ohmic and non-ohmic resistors is the relationship between voltage, current, and resistance. Ohmic resistors adhere to Ohm's law, which states that the current flowing through a resistor is directly proportional to the voltage applied across it. Non-ohmic resistors, on the other hand, exhibit nonlinear behaviour, meaning that the relationship between voltage, current, and resistance is not linear. Non-ohmic resistors are essential components in many electronic devices. Their unique properties make them ideal for a variety of applications, such as temperature sensing, light detection, and amplification.



BULLET POINT SUMMARY

- Non-ohmic resistors are resistors that do not obey Ohm's law, meaning that their resistance changes as voltage and current vary.
- Some common examples of non-ohmic resistors include light bulbs, diodes, and transistors.
- Non-ohmic resistors can exhibit a variety of interesting characteristics, such as non-constant resistance, non-reversible behaviour, and sensitivity to temperature and light.
- Non-ohmic resistors are essential components in many electronic devices, such as thermistors, LDRs, and amplifiers.

CASE STUDY

Title: Real-Life Applications of Ohmic and Non-Ohmic Conductors

Introduction: Ohmic and non-ohmic conductors are two different types of materials that exhibit different electrical properties. Ohmic conductors have a constant resistance, regardless of the voltage or current flowing through them. Non-ohmic conductors, on the other hand, have a resistance that changes depending on the voltage or current flowing through them. Ohmic and non-ohmic conductors are used in a wide variety of electronic devices and systems. Their unique electrical properties make them ideal for a variety of applications, such as controlling the flow of current, rectifying AC current, and generating heat.

Objective: The objective of this case study is to explore some real-life applications of ohmic and non-ohmic conductors.

Methodology: This case study will use a variety of sources, including research papers, textbooks, and online articles, to identify and discuss real-life applications of ohmic and non-ohmic conductors.

CASE STUDY

Findings: Here are some examples of real-life applications of ohmic and non-ohmic conductors:

Ohmic conductors:

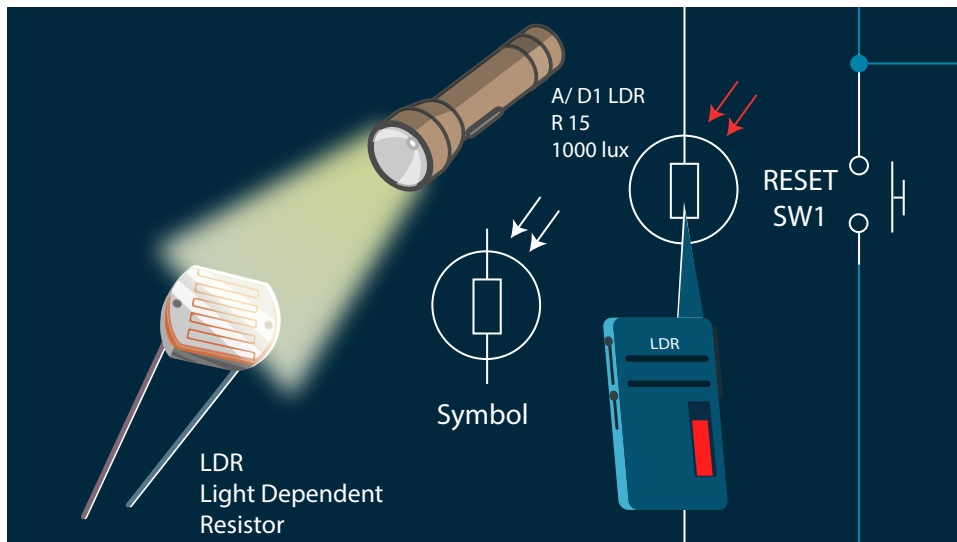
- **Resistors:** Resistors are used in a wide variety of electronic circuits to control the flow of current. They are made of ohmic materials, such as carbon, metal alloys, or semiconductors.
- **Wires:** Wires are used to conduct electricity from one place to another. They are made of ohmic materials, such as copper or aluminium.
- **Heating elements:** Heating elements are used to generate heat in appliances, such as toasters, stoves, and hair dryers. They are made of ohmic materials, such as nichrome or tungsten.

Non-ohmic conductors:

- **Light bulbs:** The resistance of a light bulb increases as it heats up. This is why light bulbs glow brighter when they are warm.
- **Diodes:** Diodes only allow current to flow in one direction. This is because the resistance of a diode is very high in one direction and very low in the other.
- **Transistors:** Transistors can act as amplifiers or switches. The resistance of a transistor depends on the input voltage and current.

QUESTIONS TO PONDER

- What are some other examples of real-life applications of ohmic and non-ohmic conductors?
- What are the advantages and disadvantages of using ohmic and non-ohmic conductors in power supplies?
- How can we design power supplies to be more efficient and reliable?



SKILLS AND COMPETENCIES YOU HAVE ACQUIRED AFTER THIS LESSON

Ability to identify and explain the nonlinear relationship between current and voltage in a non-ohmic resistor.

Ability to design and conduct experiments to study the behaviour of non-ohmic resistors.

PERSONAL REFLECTION

I learned a lot about the different types of non-ohmic resistors, their characteristics, and their applications. I also learned how to design and conduct experiments to study the behaviour of non-ohmic resistors.